

Date	15 February 2025
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	5 Marks

SMART METRO TICKET GENERATING SYSTEM

Project Planning Phase

1. Purpose of the Planning Phase

The Project Planning Phase converts the approved Problem–Solution Fit and Proposed Solution into an executable plan: a scoped, sequenced, and resourced set of activities that the team can commit to and measure progress against. For the Smart Metro Ticket Generating System, this phase translates the high-level three-phase rollout (Pilot, Regional, Network-wide) into concrete planning artifacts — a project timeline, a prioritized backlog, and clearly assigned roles — before any sprint work begins.

Good planning at this stage reduces downstream risk: ambiguous scope is the single largest cause of schedule slip in software delivery, so the planning phase deliberately front-loads the work of clarifying scope, sequencing dependencies, and agreeing on how progress will be tracked.

1.1 Planning Workflow

The diagram below shows the seven-step workflow used to move from project initiation to closure. Steps 4 through 6 (Sprint Planning, Execution, Review & Retrospective) repeat in a loop for every sprint until the product backlog is exhausted or the release scope is met.

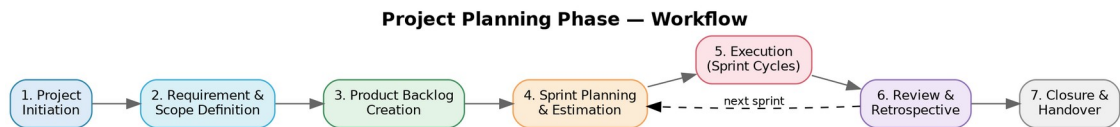


Figure 1: Project planning phase workflow, including the iterative sprint loop

1.2 Planning Logic

The planning logic follows a top-down decomposition: the overall product vision is broken into epics, epics are broken into user stories, and stories are broken into tasks small enough to estimate reliably (typically completable within 1-2 days). Prioritization at every level uses a consistent value-versus-effort lens, so that the highest-value, lowest-risk work is always scheduled first — directly mirroring the problem-severity prioritization established in the Problem–Solution Fit document.

- Decompose top-down: Vision → Epics → User Stories → Tasks, each level small enough to plan reliably.
- Prioritize by value/effort: high-impact, low-effort items move to the front of the backlog first (Phase 1 pilot capabilities).
- Plan in fixed-length sprints: a constant sprint length (2 weeks) makes velocity comparable release over release.
- Re-plan every sprint: the backlog is re-prioritized after every sprint review based on stakeholder feedback and change requests.

2. Project Timeline

The Gantt-style timeline below sequences the planning phase against the twelve weeks leading into full development, showing how requirement analysis and backlog/sprint planning overlap with early design work before development sprints begin. Testing, deployment and closure follow the same phased rollout logic established in the Proposed Solution document.

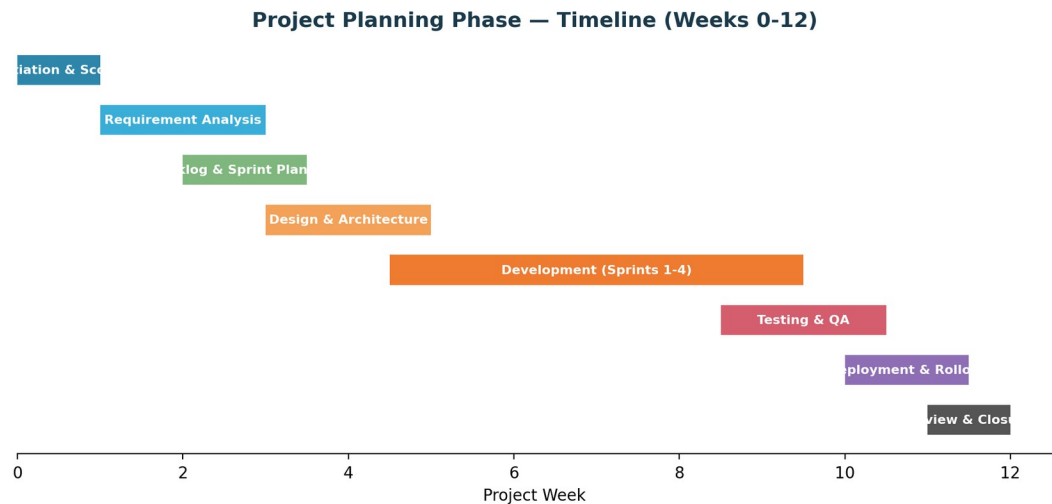


Figure 2: Project planning phase timeline across a 12-week planning-to-delivery window

2.1 Phase Breakdown

Phase	Key Activity	Primary Owner
Initiation	Define project goal, sponsors, success criteria	Product Owner
Requirement & Scope	Gather requirements, define scope boundaries	Product Owner / BA
Backlog Creation	Break scope into epics and prioritized user stories	Product Owner
Sprint Planning	Estimate, select and commit stories per sprint	Scrum Master / Team
Execution	Build, test and integrate increments each sprint	Development Team
Review & Retrospective	Demo increment, inspect and adapt process	Whole Team
Closure	Final handover, documentation, lessons learned	Project Manager

3. Roles, Responsibilities and Risk Planning

Clear ownership at each planning step avoids the most common planning failure mode: activities that everyone assumes someone else is doing. The diagram below assigns explicit ownership across the four core roles involved in planning the Smart Metro Ticket Generating System.

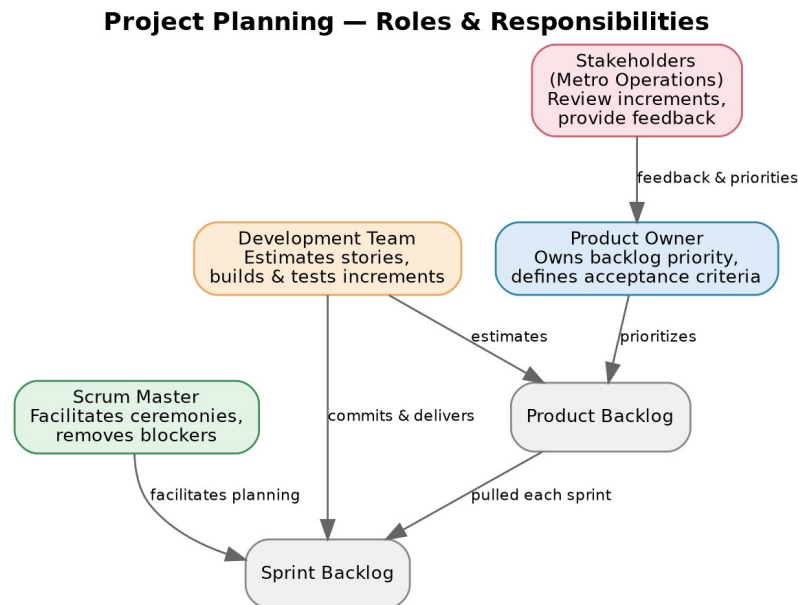


Figure 3: Roles and responsibilities across the product backlog and sprint backlog

3.1 Risk Planning

- Scope risk: an unbounded backlog is mitigated by freezing epic-level scope at the end of Initiation and only allowing new items into future sprints, not the current one.
- Estimation risk: story points are calibrated against 2-3 reference stories agreed by the whole team before the first sprint, reducing early-sprint estimation drift.
- Dependency risk: cross-team dependencies (e.g., payment gateway sandbox access) are identified during Initiation and tracked as blocking items, not discovered mid-sprint.
- Resourcing risk: the plan assumes fixed team capacity per sprint; planned leave or shared resources are accounted for before sprint commitments are made, not after.

3.2 Summary

The Project Planning Phase gives the Smart Metro Ticket Generating System a repeatable rhythm: a clear workflow from initiation to closure, a timeline that sequences planning against delivery, and named ownership for every planning activity. This phase feeds directly into the Project Planning Template — the concrete backlog, sprint plan, and story-point framework used to run each two-week sprint.